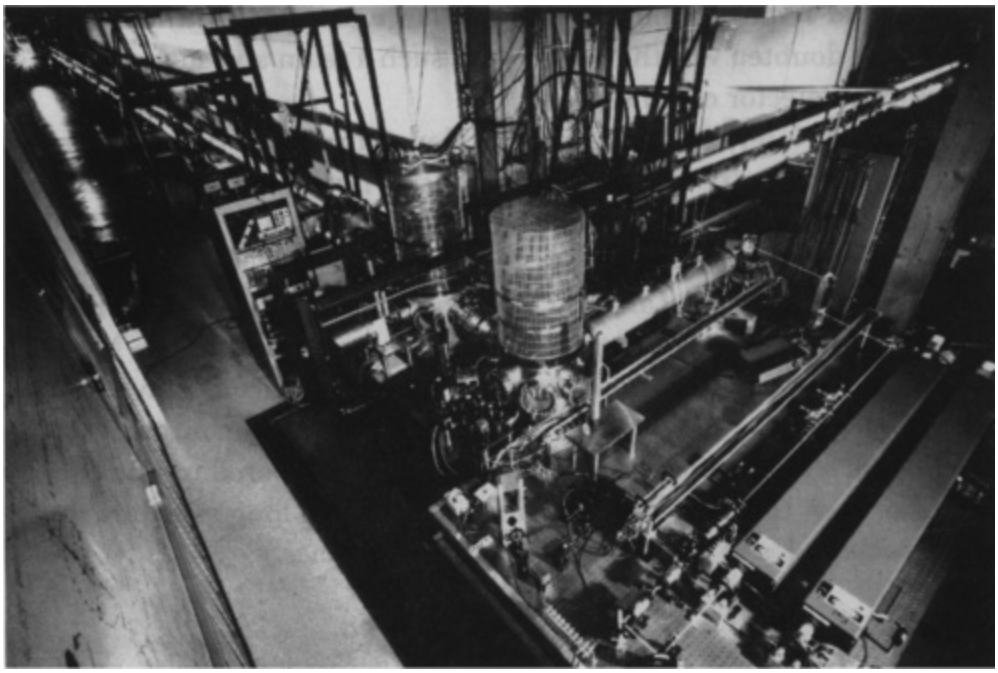


10.7 The Caltech 40-meter interferometric prototype gravitational-wave detector, ca. 1989. The table in front and the front caged vacuum chamber hold lasers and devices to prepare the laser light for entry into the interferometer. The central mass resides in the second caged vacuum chamber—the chamber above which a dangling rope can be seen faintly. The end masses are 40 meters away, down the two corridors. The two arms' laser beams shine down the larger of the two vacuum pipes that extend the lengths of the corridors. [Courtesy LIGO Project, California Institute of Technology.]

During the early 1980s four teams of experimental physicists struggled to develop tools and techniques for interferometric detectors: Drever's Caltech team, the team he had founded at Glasgow (now led by James Hough), Rainer Weiss's team at MIT, and a team founded by Hans Billing at the Max Planck Institut in Munich, Germany. The teams were small and intimate, and they worked more or less independently,[9] pursuing their own approaches to the design of interferometric detectors. Within each team the individual scientists had free rein to invent new ideas and pursue them as they wished and for as long as they wished; coordination was very loose. This is just the kind of culture that inventive scientists love and thrive on, the culture that Braginsky craves, a culture in which loners like me are happiest. But it is not a culture capable of designing, constructing, debugging, and operating large, complex scientific instruments like the several-kilometer-long interferometers required for success.